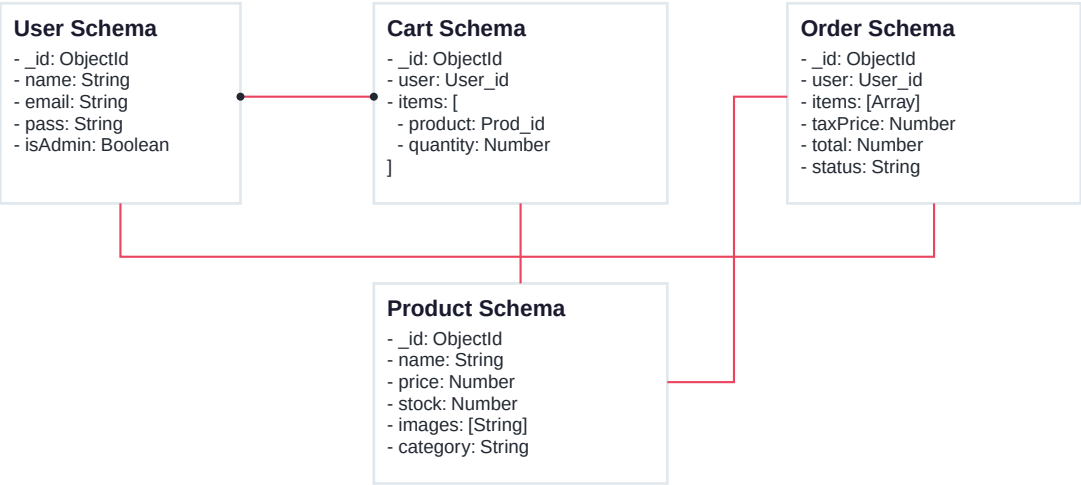

Solution Architecture & Codebase Layout

1. Backend Directory Structure

The backend utilizes a MVC-adjacent file separation structure in Express:

- server.js - Root listener initiating DB connection and endpoints.
- config/db.js - MongoDB connection module using Mongoose.
- controllers/ - Router callbacks implementing business logic.
- models/ - Mongoose schemas enforcing document formats.
- routes/ - REST endpoint pathways mapped to controllers.
- middleware/ - JWT verifications, role checks, and Multer file rules.

2. Database Entity-Relationship Diagram (ERD)



3. Essential Security Middlewares

Authentication is managed via HTTP Authorization headers containing bearer JSON Web Tokens. Admin-only controllers are protected by an admin validation middleware that blocks requests if User.isAdmin resolves to false, preventing unauthorized API queries.